Brain-Based Indicators of Passenger Trust During Open-Road Driving

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***Abstract*—Autonomous Driving has been continuously improving with the integration of bio-physiological data and deep learning systems. Passenger-driver behavior from psychological processes can provide insight into the trust levels of the dyad dynamic during an open-road drive. Measuring passenger trust can benefit in speeding up the adoption of self-driving cars. This area has not been fully explored from the passenger's point of view and can yield results that could potentially propel the progress of driver safety systems in autonomous driving. A novel approach to correlating the passenger-driver trust levels can be elicited from brain-based indicators that are extracted from time-locked electroencephalogram (EEG) signals that capture both sensory and cognitive processes. In this paper, we propose a trust identification technique utilizing evoked response potential (ERP) events such as P300 along with beta/alpha frequency ratios from specific passengers during braking, aggressive acceleration, and lane changing scenarios. The results obtained from the decomposition of these EEG signals into the frequency bands and application of machine learning techniques on data collected from frontal and parietal electrodes during these driving events prove the feasibility of this study. By examining additional passenger-driver pairs with varying social interaction and trust profiles in the future, we can strengthen the existence of a cognitive correlation between passenger-driver behavior and thus improve the efficiency of driver safety systems.**

***Index Terms*—P300, EEG, Spectral Analysis, Autonomy-Enabling Technology, ERP, Self-Driving Car, Vehicle-Passenger Behavior**

I. Introduction

In the United States, there are approximately 6 million reported car accidents annually with no noticeable improvement between 2010-2018 [1]. One would expect these statistics to improve as advancing driver safety systems become more commonplace, however with the current average car age of approximately 12 years, it seems that slow adoption rates perpetuate a circumstance wherein most consumer-grade vehicles continue to have antiquated safety systems [2]. Active safety systems are expected to offer considerable improvements, with today's most commonly available active driver safety systems including lane-keeping assist, adaptive cruise control, crash mitigation, and blind spot detection. Features that use feedback control, such as adaptive cruise, can be further categorized as an automation since the motorist is no longer required to use the brake or throttle controls unless in emergent situations. Vehicle automation has proven to be beneficial in many cases, but considerable shortfalls persist largely because of human factors issues, such as automation-induced complacency and inappropriate trust, that lead to improper system use and concomitant performance and safety reductions [3]. Ideally drivers of the near future will begin to opt for active driving automation for performance and safety enhancement, but, we argue, this will not happen broadly until we better understand how to address the human factors issues that persistently undermine adoption rates.

The self-driving car (SDC) is no longer a novel product as manufacturers have begun to quickly close the gap between themselves and the industry leader. Additionally, technology which enables complete autonomy is quickly emerging and advancing where it is expected that most of the industry will launch fully autonomous vehicles (AV) by the end of the decade [4]. However, being a passenger in an AV can induce anxiety, distrust, and discomfort similar to what a passenger experiences when riding with an unknown or inexperienced motorist [5], [6]. Public skepticism of AV inhibits its universal acceptance and creates a sense of distrust in the emerging technology, and can be attributed to several incidents involving automated vehicles, specifically SDC. However, SDC are expensive and are therefore not attainable by the public majority. Instead, an increasing number of motorists own vehicles equipped with automated features and safety systems previously mentioned. A particular cause of concern is unwarranted instances of crash mitigation, especially when driving at highway speeds. Over 400 complaints have been submitted to the National Highway Traffic and Safety Administration (NHTSA) over the last 3 years, where 14 complaints involved

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collisions and 18 involved injuries related to the sudden jerk of the brake [7]. The 14 complaints that resulted in a crash included abrupt braking that resulted in the vehicle being rear-ended, or more terrifying, an instance on the highway where the emergent braking caused the motorist to lose control of the vehicle and collide into a guardrail [7]. Aside from its shortcomings, automated features, including crash mitigation, have been reported to reduce the number and severity of roadway collisions [8]. Although less sophisticated than the technology of a SDC, the shortcomings of vehicle automation affects more motorists, and as a result, has a greater influence on public opinion.

Skepticism and public opinion have not yet hindered the demand for SDC since it is a niche market, especially when integrated with electric vehicle technology (EVT) as trail blazed by Tesla. Consumers who want to own a SDC may have a greater propensity to trust AV, and may become overtly complacent to its limitations as reported in several cases where a Tesla crashes while operating in auto-pilot. Motorists are required to be ready to take over the vehicle based on their discretion, but those who are distracted or unconscious are at risk as the auto-pilot function does not classify the SDC as fully autonomous. This misunderstanding further dampens public trust in AV.

Given the apparent centrality of a passenger's trust to their decisions regarding when and how to use a driving automation, it is easy to envision use cases for in-vehicle monitoring of a passenger's trust state. A common way to measure the degree of trust is using self-report in the form of responses to survey questions; unfortunately, the degree of subjectivity and inconsistency involved in human self-reports of subjectively-experienced states taken in combination with the abysmal temporal resolution of such data points demands alternative approaches. The present work explores the application that human brain-based data may provide an important window into the trust-based decision processes that we believe underlie automation use behaviors [9].

II. Background

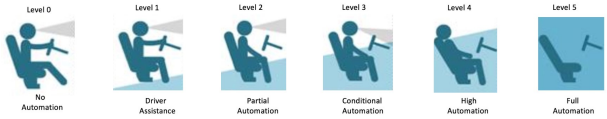

Fig. 1. Automation Tier of Autonomous Vehicles experiment

The Society of Automotive Engineers (SAE) describes 6 levels of automation for AV as illustrated in Figure 1. Level 0 consists of features that provide momentary assistance like automatic braking and blind spot detection. Whereas Level 5 is characterized by complete autonomy with the ability to perform under any condition and requires no passenger-vehicle interaction. [10]. In Level 5 automation, the passenger can take over the vehicle if need be, but the purpose of a fully AV is to eliminate the need for passenger-vehicle interaction without fear of compromising the safety of the passenger, pedestrians, or other motorists [11]. By having a high propensity of trust in level 5 AV, the passenger will be less likely to take over the vehicle, reducing the associated risks of human error.

Quantifying trust and other emotional behavior is complex but integral in the development of fully autonomous vehicles. Each human carries a rich array of knowledge and experiences into their daily activities, which greatly influences their decision making patterns and behavioral styles. If AV were enabled with the intelligence to perceive and respond to the subtle changes in emotional and behavioral patterns, they can ascertain the passengers perception of risky driving and adapt accordingly. In doing so, the dynamics between the passenger and intelligent AV are optimized and results in greater passenger trust. However, research in passenger-vehicle interactions in AV is limited due to varying state regulations and commercial availability. Additionally, experimentation in a secured and controlled setting with AV or virtual simulation reduces the reality of risk and implicates meaningful analysis. Taking this into consideration, an equivalent if not better analysis of trust can be conducted through passenger-driver interactions on open roadways. In this scenario, the passenger acts as a proxy to being in a level 5 AV and the naturalistic drive provides the perception of realistic risk. In both cases, the role of the passenger does not change as they have little to no control over the drive.

Trust can be evaluated during risky activity by indicators for emotional response evident in spectral analysis of event-related potentials (ERP). The Event-Related Potential of the P300 wave is identified between 250-500 ms after a stimulus has been perceived, but can vary based upon subject age, task type, and stimulus modality (i.e visual versus auditory) [12]. The P300 characterizes cognition but can also be decomposed as P3a and P3b waves. The P3a wave propagates from the frontal lobe during task processing while the P3b propagates from the temporal-parietal region during tasks or events that increase attention [12]. The ERP can be analyzed based upon latency and amplitude and is best suited for response and inhibition studies where reaction time is emphasized but becomes more robust when transformed into the frequency domain as delineated in [13] and [14] where emotional and cognitive states were identified via machine learning. Additionally, ERPs can be processed using statistical methods such as linear discriminate analysis (LDA) for classification. For example, the ERP can be classified as safe or unsafe driving via LDA [15].

Higher frequency bands have been reported to play a critical role in assessing emotional response. The Alpha (8-13 Hz) and Beta (13-32 Hz) bands are commonly associated with relaxation and alertness and were shown to have a greater emotional response than the lower frequency bands [16]. These bands can be used as a power ratio to quantify positive or negative emotional state in passengers [17]. Additionally, a decrease in the alpha band and an increase in the beta band has been reported as an indication of stress and can potentially be used as an index of trust [18].

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To quantify the magnitude of each frequency band of the ERP, the ERP must first be transformed from its time domain into the frequency domain by Discrete Fourier Transform (DFT). Next, the Power Spectrum Density (PSD) must be calculated, filtered by frequency, and then averaged to produce one value for each frequency band. Alternatively, the Welch Method can be executed to obtain the power spectrum [17].

III. Method

The objective of this study is to explore the application of EEG-based measures on data recorded from three individuals reported to have high, low, and average trust, and to present changes in power spectra to gauge emotional response during real-world driving events such as braking, accelerating, and lane changing. Common techniques utilize the quantification of emotional state through a beta/alpha power ratio where a high ratio indicates a negative emotional response, and a low ratio indicates a positive response [17]. These frequency bands are of interest for this study, but only for the electrodes placed over the frontal and parietal lobes as P300 waves are known to originate there [12].

A. Dataset Description

The data set analyzed in this study was obtained by the US Army Research Laboratory and re-purposed for further analysis outside of the original research goals. The data was collected on the open roadway of I-95 near Aberdeen, Maryland and was conducted in a 2016 Ford Fusion Titanium which is equipped with a suite of driver safety features, however, these were disabled due to inexperience in the participant majority. The route is approximately 24.5 miles long and is shown in Figure 2. The route begins and ends at the same home point and composes of both urban and highway roadways where approximately 18 minutes of the total 25 minute one-way drive was conducted on I-95. The study included up to 53 subjects and consisted of either male or female, civilian and contractor employees at ARL. Drives were designed in a pair known as a dyad where both participants repeated the drive once as the driver and once as the passenger. The total participation time was 2.5 – 3 hours which includes the 50-60 minute drive time as well as the before and after procedures. Participant risk due to unforeseen roadway encounters was mitigated by avoiding times of roadway congestion and adverse weather, and by optimizing passenger safety. Additionally, a safety operative was included and seated in the rear of the car to aid with initializing the system as well as troubleshooting the equipment to reduce stress placed on the participants. By mitigating passenger risk without overtly affecting perceived risk, the procedure was compliant with regulations set forth by the U.S. Federal Law and U.S. Army as well as those set by the Institutional Review Board (IRB).

Data was collected using a Multi-Aspect Real-world Integrated Imaging (MARIN) system designed and coined by ARL which consists of synchronized commercial off-the-shelf technology that enables monitoring of pyscho-physiological, behavioral, and subjective tasks in real-world scenarios. The

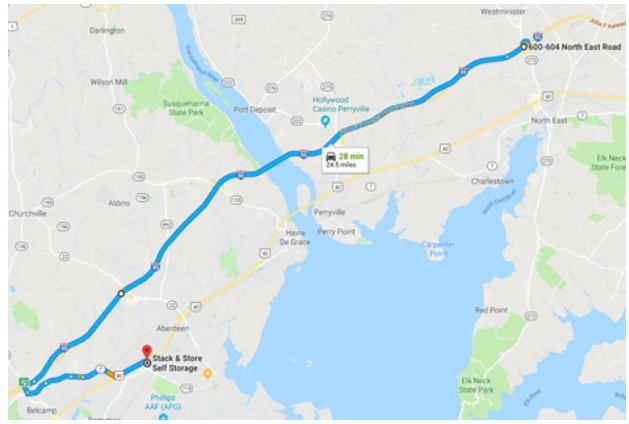

Fig. 2. *R*oute of I-95 for the conducted experiment

MARIN system can log electroencephalographic (EEG), electrocardiographic (ECG), electrodermal (EDA), photoplethysmography (PPG), respiratory, and physical activity, and skin temperature. The neural component of the MARIN system is the ABM B-Alert x24 which produces an EEG over 20 electrodes as illustrated below in Figure 3, however, it can also produce electrocardiogram, electrooculogram (EOG), and electromyogram (EMG) over its 4 remaining electrodes with 2 being references. Of these 20 Electrodes, 10 were chosen over the frontal and parietal regions of the brain as previously explained - F3, F1, Fz, F2, F4, P3, P1, Pz, P2, P4.

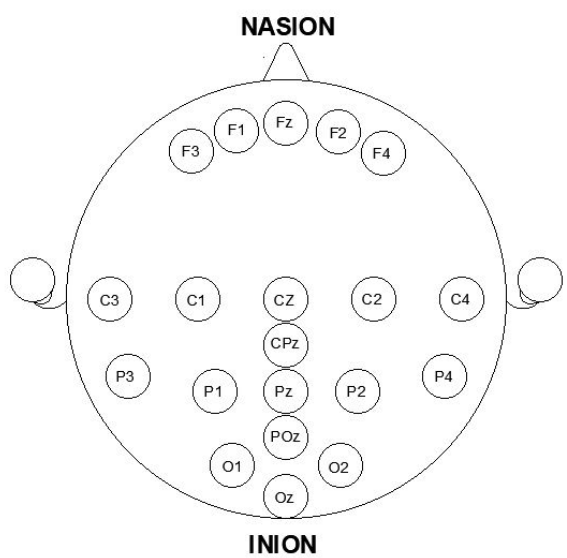

Fig. 3. *E*lectrode positioning map of the ABM B-Alert x24 EEG

For the purposes of the present work, 3 of the 43 collected sessions will be processed as a preliminary report on the feasibility of applying methods for both event-related and spectral analyses on EEG data recorded during real, unconstrained driving on the open roadway. The three sessions chosen were based on levels of trust. Specifically, EEG data for a passenger

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with average, low and high passenger-driver-trust behavior was chosen. Provided positive outcomes, the indicators explored here will be leveraged into a multi-modal analysis along with other classifiers in collaboration with Kennesaw State University (KSU) and ARL.

B. Participant Tasks

Before each drive, the driver and passenger were subject to a viewing activity that was tablet-based. The driver received briefing on the route, the safety features of the car, and its controls, while the passenger received video stimuli that composed of news items. The drive itself was on a 50-minute course and comprised of highway roads and a local city. Once the drive began the passenger was motioned to talk about the news stimuli, while the driver focused on prioritizing safe driving. The driver was instructed to listen passively. Following the drive, the driver was instructed to report back memories of the information remembered during the drive. Each participant also completed a survey both before and after the drive.

C. EEG Signal Conditioning

Due to the sensitive nature of the EEG, it can easily become polluted by artifact created by other electrical bio-signals. Additionally, unavoidable head movement can compromise the connection of the electrodes to the scalp as well as introduce artifact noise via muscular activity. Initially, the EEG was synchronized with other physiological signals collected during the drive, and eventually conditioned through line noise removal, bad channel detection and interpolation, and channel re-referencing in a procedure known as the PREP Pipeline thoroughly delineated in [19].

D. EEG Signal Processing

This paper focuses on the EEG data collected form three passengers with high, low, and average trust in the driver. Before ERP analysis, the event file (.tsv) was scanned for exact strings matching the event type (braking, accelerating, and lane changing) then their times were stored in an array. Due to discontinuities in time between the EEG file (.csv) and the event file, exact matches were a challenge to obtain. To resolve this issue, the event time array was transformed into an index value nearest to the desired time in the EEG file. Now that the event time is correlated with the EEG data, MNE-Python package, an open source application, can be utilized for simplified analysis, but first, epochs for each event must be defined. To capture the P300 wave in our analysis, epochs were windowed 200-500 ms after the event time. From there, epochs were averaged and concatenated into a single plot for each event type. Lastly, the Welch method for power spectra was performed on the epochs to obtain the absolute and relative power of each frequency band. Since the Welch method for power spectral density reduces noise at the cost of reduced frequency resolution, it was verified, as seen in Figure 4, and preferred over other methods since EEG is prone to noise.

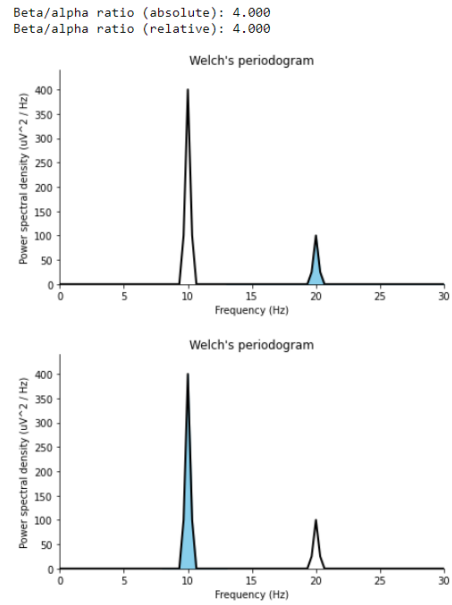

Fig. 4. *V*erification of the Welch Method using Composite Sinusoidal Input

IV. Results

The events that were analyzed were driver starts braking, aggressive acceleration, and lane shifts. In each trust relationship, the number of braking, aggressive acceleration, and lane changes were tabulated. There were 32 braking events, 72 aggressive acceleration events, and 12 lane changes for the average trust session. Similarly, the study captured 60 and 63 braking events for the low and high trust sessions. The events in each trust session were analyzed and compared to show a cognitive relationship between the trust profiles and human factors. The number of events in each session is shown in Table I.

TABLE I
EVENT COUNTS

Event	**Avg. Trust**	**Low Trust**	**High Trust**
1. Driver Starts Braking	32	60	63
2. Aggressive Acceleration	72	90	52
3. Lane Changes	12	33	17

The machine learning functionality of the algorithm successfully identified the average event related evoked potential wave P3 for each passenger trust session as shown in Figures 5, 6, & 7. The x-axis represents the time series in seconds, while the y-axis represents the voltage magnitude (V) of the wave. Figure 5 shows the ERP plot for average trust dyad profile beginning with braking events, and followed by aggressive acceleration, and lane changes events.

The ERP analysis of EEG waves for the frontal and parietal channels from these events indicated P3 waves in the EEG signal. The P3 peaks in the low trust profile indicated high level of EEG activity in all three events - driver starts braking, aggressive acceleration and lane changes displayed in Figure

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6. The P3 plots for the high trust profile indicated a moderate level of EEG activity for the same events displayed in Figure 7. On the other hand, the P3 plots for average trust profile indicated an average level of EEG activity. The three passenger-driver sessions successfully indicated P300 peaks in the EEG signal. The analysis of the resulting signals demonstrates a correlation between the subjective trust relationship of passenger and driver to the spikes in EEG signals.

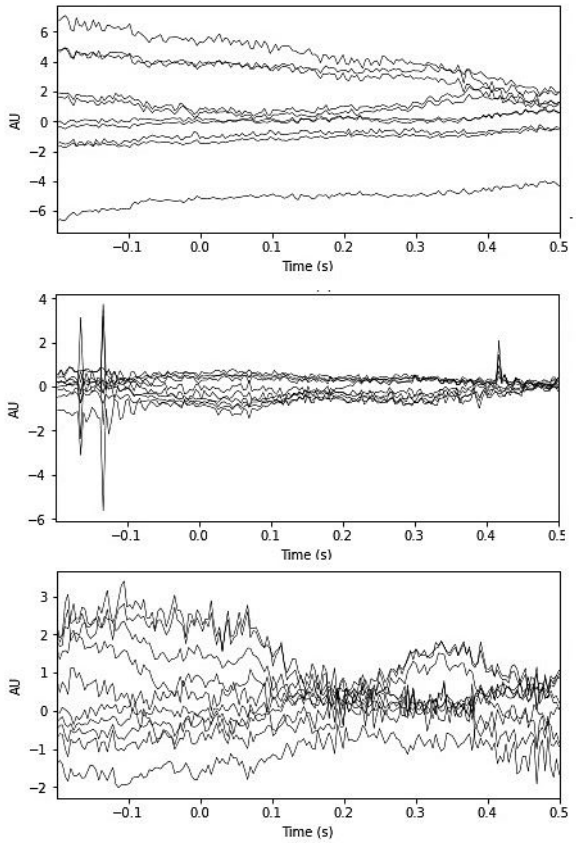

Fig. 5. Average Trust- Driver Starts Braking, Aggressive Acceleration, and Lane Changes (top to bottom)

The results were in line with the expectation of average evoked responses and increased P3 waves at the onset of the three chosen driving events. In addition, events modeled around the driver braking showed a higher potential of trust prediction, whereas events modeled around the driver changing lanes showed a lower evoked response. A reason for this could be the smaller number of events in the lane changing event for the three profiles. In the average trust profile, the number of lane change events was small. Although the study grouped left and right lane changes into one group in an attempt to increase the number of observations, the number of data points was still too small to produce a significant evoked response in the machine learning algorithm. In future studies, it could be worthwhile to specifically study left and right lane change events separately in order to determine further relationships between passenger-driver behavior and

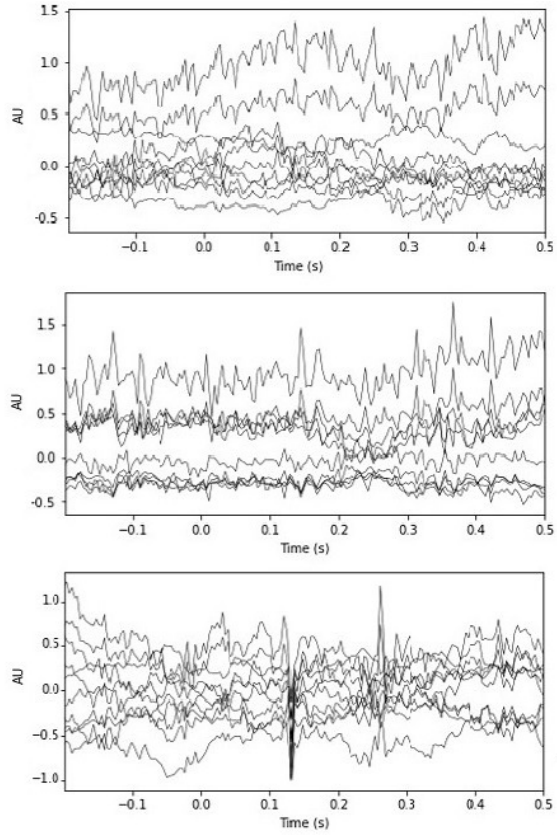

Fig. 6. Low Trust- Driver Starts Braking, Aggressive Acceleration, and Lane Changes (top to bottom)

brain based indicators. This would be beneficial if the number of lane changes was significant in an open drive. Another observation was that aggressive acceleration event produced significant evoked response events in both the low and high trust profiles. The qualitative analysis of the results explores the feasibility of further study in similar directions.

V. Conclusion

The purpose of the research study was to highlight the feasibility of studying EEG waves from specific driving events in relation to brain-based indicators and passenger-driver trust. The results from plotting the P3 wave in the EEG signals showed that patterns in event related potentials can be good indicators of the level of passenger-driver trust. Patterns in the frequency band beta/alpha ratios can also be studied from various passenger-driver trust scenarios. This novel approach to correlate passenger-driver trust behavior with brain-based indicators satisfied the purpose of the research. These indicators can be incorporated into driver safety systems to predict risky driving conditions. Future analysis of EEG data from additional passenger-driver pairs and their trust levels can produce improved results and is justified. Future research work includes studying frequency band analysis. The results are important and can be applied to further autonomous

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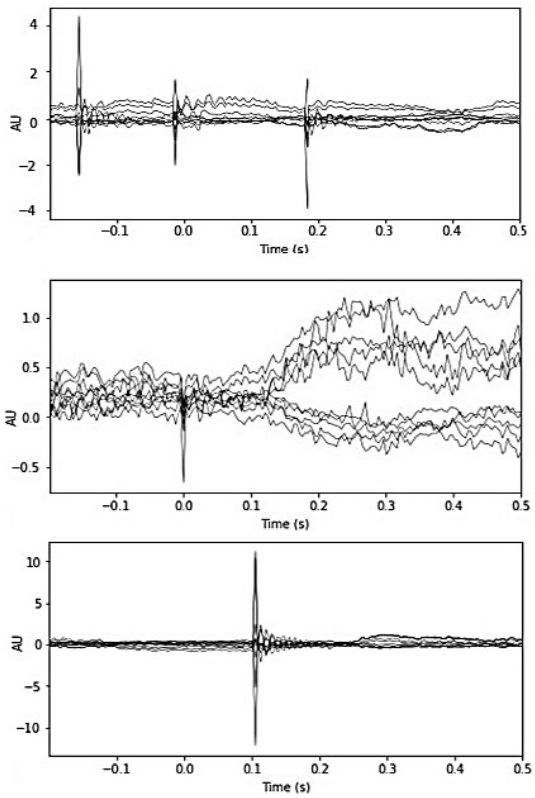

Fig. 7. High Trust-Driver Starts Braking, Aggressive Acceleration, and Lane Changes (top to bottom)

driving studies. They can be adopted in assisting the future development of intelligent systems to predict risky situations in open-road driving. Variations in social indicators of the passenger such as familiarity of the driver, risk profiles of the driver, attentiveness of the passenger and so on can produce additional insights on the trust results. The study can be extended to include additional events such as using lane blinkers, accelerating at a normal rate, decelerating aggressively, and speed variations. The results from related events can provide motivation for future research direction and progress in the development of autonomous intelligent systems.

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